AGENTIC AI SUMMIT 2026 · UC BERKELEY

Small models. Tight loops.

What a coding-agent harness is actually doing for you

one Python file | real compiler | external postconditions

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```
nanagent / run 01

$ askme "build a two-file C program"

plan   write .h → write .c → compile → run
tool   cc -o main main.c
check   ./main

AGENT_OK
```

A REAL FAILURE TRACE

The bug that ate minutes

```
// missing: #include <stdio.h>
int main() {
  printf("FIXED\n");
  return 0;
}

error: implicit declaration of printf
```

step 1 — **Compile**

Useful failure: the compiler names the bug.

next — **Bad edit**

Malformed JSON or the wrong exact-match string.

140–253s — **Think, retry, re-read**

The model spends tokens rediscovering known state.

Wrong move: buy more thinking. **Better move:** turn the compiler error into control flow.

Three traces of the slowest local microtask, Apr 26, 2026 · PERFORMANCE.md. The 140–253s range is not a suite-wide result.

HARNESS DESIGN

Move reliability out of the model

01

Typed error

`compile_error`, not an undifferentiated failure string.

→

02

Exact context

Read the file before retrying an edit.

→

03

Cheap repair

Use a narrow deterministic fix when the diagnostic is unambiguous.

→

04

Small replan

Replace one failed task, not the whole plan.

→

05

Real check

Run the artifact outside the LLM's own judgment.

```
compiler stderr → [compile_error] → repair / retry → ./main == "AGENT_OK"
```

Implementation: askme.py · architecture and historical timings: ARCHITECTURE.md, PERFORMANCE.md

MEASURED OPENROUTER SMOKE · JULY 10, 2026

Four models, two concrete jobs

MULTI-FILE BUILD

Header + source → compile → run

`msg.h + main.c · ./main == REPLAN_OK`

REPAIR

Fix a Python syntax error

`python3 greet.py exits 0 · stdout contains hello`

Gemma 4 26B A4B	Gemma 4 31B	Qwen3.6-27B	Qwen3.6-35B-A3B
MoE · 3.8B active	dense · 30.7B	dense · 27B	MoE · 3B active

Held fixed: NanAgent implementation, task prompt, SiliconFlow-only routing, one retry policy, strict external checks. Unique catalog matches were FP8. **n=1/cell.** 31B was a post-hoc extension.

Exact protocol and commands: evals/README.md · Model architecture: official Google and Qwen model cards

EIGHT MEASURED CELLS · JULY 10, 2026

What happened?

Model	2-file build	Repair	Responses	Billed credits
Gemma 4 26B A4B	**PASS** 603.6s	**PASS** 20.0s	39	$0.004147
Gemma 4 31B	**PASS** 66.5s	**PASS** 22.4s	19	$0.001321
Qwen3.6-27B	**PASS** 47.9s	**PASS** 23.0s	22	$0.004968
Qwen3.6-35B-A3B	**FAIL** 17.7s	**PASS** 11.8s	14	$0.001875

Pass requires pytest, **agent complete**, and the external check. “Responses” means usage-bearing model replies; raw HTTP attempts were not instrumented.

7/8 passed. Gemma: both 2/2 · 31B build used 8 vs 29 responses · repair used 11 vs 10. Size + architecture changed.

Raw summary: evals/draft-results.json · route: response metadata · FP8: catalog match · infrastructure failures excluded

SCOPE CHECK

Useful evidence, small claim

What we actually verify

- Structured actions survive the model/provider path
- The loop can build or repair these exact programs
- The resulting artifact executes correctly
- Responses, tokens, route, and cost are auditable

What we do not claim

- Full-app or long-horizon reliability
- Local-laptop speed from hosted runs
- UX, architecture, or maintainability quality
- Model size alone caused the Gemma gap
- Contest-code scores predict agent performance

No LiveCodeBench claim: short contest problems answer a different question from multi-step, multi-file agent work.

TAKEAWAY

The loop is the product

Small models do not need easier standards.
They need tighter feedback loops.

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data